Protocols - Immobilised Metal Affinity Chromatography

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

*This protocol purifies two His-tagged enzymes, **DHAK** and **PPK2**, from clarified cell-free extract (CFE) by immobilised metal affinity chromatography (IMAC) on Co^{2+}-IDA-agarose. It uses a gravity-flow SPE cartridge. The procedure is identical for both enzymes; only the buffer composition differs (NaCl and imidazole), to suit how each protein tolerates salt and imidazole (see Buffer preparation). One packed cartridge processes one CFE sample and gives an eluate enriched in the target enzyme. The eluate is then dialysed (Protocol_Dialysis_of_IMAC_Eluates) to remove imidazole and excess salt before the enzyme activity assays. Compared with the starting CFE, the purified fraction should show a higher specific activity (U/mg protein) and a dominant band at the expected molecular weight of DHAK or PPK2 on SDS-PAGE.*

Materials list

Reagents

- Clarified CFE containing His-tagged DHAK or PPK2.
- Co^{2+}-IDA-agarose resin, supplied as a suspension.
- Sodium phosphate, dibasic: $Na_2HPO_4 \cdot 2H_2O$ (177.99 g/mol).
- Sodium chloride, NaCl (58.44 g/mol).
- Imidazole (68.08 g/mol).
- HCl and NaOH (1 M), for pH adjustment.
- Ultrapure water (Milli-Q).

Consumables

- Empty polypropylene SPE cartridge with bottom frit, bottom Luer tip cap and top cap/adaptor with a Luer port.
- 10–20 mL Luer syringe and a needle or Luer adaptor that fits the top cap (used to apply positive air pressure).

Equipment

- Retort stand with clamp, to hold the cartridge vertical.

Steps

A. Buffer preparation:

All buffers are based on 50 mM sodium phosphate, pH 8.0:

DHAK buffers -

- **Equilibration Buffer:** 50 mM sodium phosphate, pH 8.0
- **Wash Buffer:** 50 mM sodium phosphate, pH 8.0
- **Elution Buffer:** 50 mM sodium phosphate, 1M imidazole, pH 8.0

PPK2 buffers -

- **Equilibration Buffer:** 50 mM sodium phosphate, 150 mM NaCl, 30 mM imidazole, pH 8.0.
- **Wash Buffer:** 50 mM sodium phosphate, 150 mM NaCl, 30 mM imidazole, pH 8.0.
- **Elution Buffer:** 50 mM sodium phosphate, 1 M imidazole, pH 8.0.

Table 1. Imidazole and NaCl concentration in the IMAC buffers for DHAK and PPK2.

Buffer	DHAK		PPK2	
Equilibration	0mM imidazole	0mM NaCl	30mM imidazole	150mM NaCl
Wash	0mM imidazole		30mM imidazole	
Elution	1M imidazole		1M imidazole	0mM NaCl

Note: Adjust the pH of each buffer **after** all components (including imidazole) have dissolved and **before** making up to the final volume, as imidazole is basic and raises the pH.

B. Column assembly and packing:

1. Clamp the empty SPE cartridge vertically on the retort stand, with a waste tube underneath.
2. Rinse the cartridge with 10–15 mL of Milli-Q water and let it drain by gravity.
3. Cap the column outlet before adding the resin. ⚠️
4. Homogenise the Co^{2+}-IDA-agarose suspension by gentle inversion.
5. Load ~4 mL into the cartridge to obtain a bed of ~1.2 cm height × 0.7 cm diameter.
6. Allow the resin to settle for 1–2 min. Remove the cap and let the storage solution drain.
7. Wash the resin with at least 5 column volumes (CV) of Milli-Q water to remove the storage solution.
8. Equilibrate the resin with at least 5 CV of Equilibration Buffer (DHAK or PPK2, see section A).

Never let the resin bed run dry. Always leave a thin layer of liquid above the bed before adding the next solution. ⚠️

Optional: *to speed up the washes, apply gentle positive pressure with an air-filled syringe connected to the top cap. Stop before the liquid reaches the resin bed and never pull the plunger back.*

C. Sample loading and batch binding

1. Cap the column outlet before loading the sample, otherwise the protein will be lost. ⚠️
2. Set aside an aliquot of the CFE ***(Load, L)*** for SDS-PAGE and protein quantification.
3. Load the CFE onto the equilibrated resin.
4. Resuspend the resin in the sample by gently pipetting up and down (or gently inverting the capped cartridge) to maximise contact between resin and protein.
5. Incubate for 5 min.
6. Remove the bottom cap and collect the unbound fraction ***(Flow-through, FT).***

D. Washing

1. Cap the column outlet. ⚠️
2. Add at least 5 CV of Wash Buffer (DHAK or PPK2, see section A) and let it drain by gravity.
3. Remove the cap and collect the Wash fraction ***(Wash, W)*** without letting the resin run dry.

E. Elution

1. Cap the column outlet. ⚠️

2. Add Elution Buffer (DHAK or PPK2, see section A) in a volume no greater than that of the sample loaded (e.g. ≤ 5 mL for a 5 mL CFE sample; we typically used 2-3mL), so that the protein is not diluted.
3. Resuspend the resin gently in the Elution Buffer and incubate for ~5 min.
4. Remove the cap and collect the eluate ***(Elution, E).***
5. Keep all fractions ***(Load, Flow-through, Wash, Eluate)*** on ice for SDS-PAGE analysis and proceed with the eluate to dialysis.

Troubleshooting

- **Sample loss through the column outlet:** In our first run, the protein was lost because the outlet was not capped before loading, and the purification had to be repeated. The protocol now marks with ⚠️ every point where the outlet must be capped: before adding the resin, before loading the sample, before the wash and before the elution. Check that the cap is tight before adding any solution you need to keep.

- **Loss of activity or protein after purification:** Analyse every fraction (Load, Flow-through, Wash and Eluate) by SDS-PAGE to find where the protein is lost. Compare **specific activity** (U/mg), not total activity: total activity always drops after purification, while specific activity should rise. If activity is still lost, consider changing the purification method.

- **Dilute eluate:** The larger the elution volume, the more diluted the protein. Never use more Elution Buffer than the volume of sample loaded. If the eluate is still too dilute, reduce the elution volume further to obtain a more concentrated single fraction.